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**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT AND ENGINEERING**

**SCHOOL OF TECHNOLOGY MANAGEMENT AND ENGINNERING**

Academic Year: 2024-2025

Program/s: B Tech / MBA Tech / BTI — Year: II/III Semester: III/V

Stream/s ; COMP/EXTC/DS/CSE DS/MXTC/CS

Subject: Management Accounting for Engineers — Time:3 hrs (10 to 1) am pm

Date: 03/ 12 / 2024 — No. of Pages: 5

Marks: 100

**Final Examination (2024 - 25)**

**Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.**

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-2 ; SO- ; BL- Understand</td><td>a.</td><td>a) Identify the types of cost as under: (Under both category-whether it is direct/indirect and fixed/variable))
<table>
<tr><th>Cost</th><th>Direct / Indirect</th><th>Fixed / Variable</th></tr>
<tr><td>Salary of Production Supervisor</td><td></td><td></td></tr>
<tr><td>Materials consumed (silk) for production (silk sarees)</td><td></td><td></td></tr>
<tr><td>Wages for labourers who work in the factory</td><td></td><td></td></tr>
<tr><td>Salaries of Office Staff & Managers</td><td></td><td></td></tr>
<tr><td>Vehicle expenses used for transport of workers (factory-home-factory)</td><td></td><td></td></tr>
</table></td><td>5</td></tr>
<tr><td>CO- 3; SO- ; BL-Apply</td><td>b.</td><td>The fixed costs for the year are Rs. 600,000. The selling price per unit is Rs. 150, and the variable cost per unit is Rs. 100. The firm sold 15000 units during the year.<br>A) Calculate the break-even point in units and in rupees<br>B) Calculate the Margin of Safety</td><td>5</td></tr>
<tr><td>CO- 4; SO- ;</td><td>c.</td><td>To produce 10 units of Finnish goods, raw materials of 20 kg are required at Rs.2.25 per kg. During an accounting period 10 units are manufactured by</td><td>5</td></tr>
</table>

| | | | |
|---|---|---|---|
| BL-Apply | | employing 25 kg of raw materials at the rate of Rs.2.00. Calculate Material price variance and Material usage variance. | |
| CO-1 ; SO- ; BL- Understand | d. | "Managerial Accounting helps managers to take strategic decisions in the firm." Justify the above statement. | 5 |
| **Q2** CO-3; SO- ; BL- Analyze | a | A company requires components beta for manufacturing a product. The following are the particulars for the year.<br>**1.** Animal demand for beta 32,000 units.<br>**2.** Cost of placing an order Rs.400 per order<br>**3.** Cost per unit of beta Rs.800<br>**4.** Cost of carrying, 20% per annum.<br>Discount of 4% on beta has been offered to the company for an order size of 4000 components at a time<br>a) Calculate the economic order quantity, total cost, ordering cost, carrying cost for both at actual price and at discounted price<br>b) Advise whether the company accepted the discounted offer. | 15 |
| **Q2** CO-1; SO- ; BL- Understand | b | 'Accounting is the language of businesses, and it helps to prepare financial reports of a business, which many stakeholders are interested in. As an Engineering student, how would you substantiate the importance of accounting for engineers? | 5 |
| **Q3** CO-3; SO- ; BL- Analyze | | A company annually manufactures and sells 20000 units of a product, the selling price of which is Rs. 50 and the profit earned is Rs. 10 per unit.<br><br>The analysis of the cost of 20,000 units is<br>Material cost Rs. 300000<br>Labour cost Rs. 100000<br>Overhead (50% variable) Rs. 400000<br>You are required to compute<br>a) Contribution per unit<br>b) P/V Ratio<br>c) Break even sales in Rs.<br>d) Break even sales in units<br>e) Sales required to earn a profit of Rs. 300000<br>f) Profit when sales are 18000 units<br>g) Margin of safety when actual sales are Rs. 600000<br>h) Analyze the impact of the following changes on P.V. ratio, break-even point in each of the following cases: (Do each point individually)<br>(i) Decrease of 10% in selling price.<br>(ii) Increase of 10% in Variable costs. | [20] |

<table>
<tr>
<td>Q4<br>CO-2; SO-<br>;<br>BL-Apply</td>
<td></td>
<td>From the following particulars, prepare a Cost Statement showing the components of Total Cost and Profit for the year ended 31st March, 2024.
<table>
<tr><td></td><td>Rs.</td><td></td><td>Rs.</td></tr>
<tr><td>Purchase of Raw Material</td><td>4,75,000</td><td>Sales for the year</td><td>8,60,000</td></tr>
<tr><td>Carriage Inwards</td><td>12,500</td><td>Selling Expenses</td><td>9,250</td></tr>
<tr><td>Wages</td><td>1,75,000</td><td>Opening Stock of Finished Goods (as on 1-4-2023)</td><td>6,000</td></tr>
<tr><td>Works Manager's Salary</td><td>30,000</td><td>Closing Stock of Finished Goods (as on 31-3-2024)</td><td>15,000</td></tr>
<tr><td>Factory Employees' Salaries</td><td>60,000</td><td>Opening Stock of Raw Materials (as on 1-4-2023)</td><td>40,000</td></tr>
<tr><td>Factory Rent , taxes and insurance</td><td>7,250</td><td>Closing Stock of Raw Materials (as on 31-3-2024)</td><td>50,000</td></tr>
<tr><td>Power Expenses</td><td>9,500</td><td>Opening Stock of Work-in-progress (as on 1-4-2023)</td><td>15,000</td></tr>
<tr><td>Other Production Expenses</td><td>43,000</td><td>Closing Stock of Work-in-progress (as on 31-3-2024)</td><td>10,000</td></tr>
<tr><td>General Expenses</td><td>32,500</td><td></td><td></td></tr>
</table>
</td>
<td>[20]</td>
</tr>
<tr>
<td>Q5<br>CO-3; SO-<br>;<br>BL-<br>Analyze</td>
<td>a</td>
<td>A firm sells 25000 units at a selling price of Rs. 5 per unit. Its fixed cost is Rs. 40,000 and variable expenses Rs. 50,000. Find out the break even point for the firm. Also, find out BEP in units in each of the cases:<br>1) The selling price is increased by 30%<br>2) The fixed cost is increased by 15%<br>3) The fixed cost is decreased by 25%<br>4) The selling price is deceased by 20%</td>
<td>[12]</td>
</tr>
<tr>
<td>Q5<br>CO-3; SO-<br>;<br>BL-<br>Understa<br>nd</td>
<td>b</td>
<td>Explain Budgeting along with the advantages a company may get from Budgeting process.</td>
<td>[8]</td>
</tr>
<tr>
<td>Q6</td>
<td>a.</td>
<td>Materials are the most important element of variable cost of a production process. Therefore, material control is a headache for the management. However,</td>
<td>[8]</td>
</tr>
</table>

<table>
<tr><td>CO-2; SO-;<br>BL-<br>Understa<br>nd</td><td></td><td>there are multiple material control techniques available at present. Explain the following techniques of material control.<br>A) ABC<br>B) FSN</td><td></td></tr>
<tr><td>Q6<br>CO-5; SO-;<br>BL-<br>Create</td><td>b.</td><td>A manufacturing company is operating at 75% of normal capacity. Normal capacity is 1,00,000 units. It is proposed to offer a price reduction of 5% to 10% depending upon the sales volume desired. Given below are the relevant data.<br><br><table><tr><td>Capacity</td><td>75%</td><td>85%</td><td>100%</td></tr><tr><td>Output (units)</td><td>75,000</td><td>85,000</td><td>1,00,000</td></tr><tr><td>Selling price per unit</td><td>₹ 96</td><td>5% off</td><td>10% off</td></tr><tr><td>Material cost per unit</td><td>₹40</td><td>10% less</td><td>15% less</td></tr><tr><td>Wages cost per unit</td><td>₹ 10</td><td>₹ 10</td><td>₹10</td></tr></table><br><u>Fixed overheads:</u><br>Production ₹14,00,000<br>Selling and administration ₹5,00,000<br><u>Variable overhead:</u><br>Production ₹ 14,00,000 @ normal capacity<br>Selling and administration ₹ 4,40,000 @ normal capacity<br>(a) Prepare a single statement to show profit/Loss at each level of output.<br>(b) Compute unit variable cost, unit fixed cost and unit total cost at different levels of output and<br>(c) Indicate which of the 3 levels is most profitable.</td><td>[12]</td></tr>
<tr><td>Q7<br>CO-2; SO-;<br>BL-Apply</td><td>a.</td><td>For the manufacture of a product X, two components A and B are used. The following particulars about the components are<br>Normal usage for A and B (per week) 300 nos<br>Maximum usage of A and B (per week) 450 nos.<br>Minimum usage of A and B (per week) 150 nos.<br>Reorder Quantity A – 2400 nos,<br>B – 3600 nos.</td><td>[8]</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | Reorder period A- 4 to 6 weeks,<br>B- 2 to 4 weeks.<br>You are required to calculate the following for both the components.<br>Reorder Quantity<br>Maximum level<br>Minimum level<br>Average level | |
| **Q7**<br>CO-4; SO-<br>;<br>BL-Apply | b. | The following are the details for P Ltd for the month of Jan 2023.<br>The Standard labour hours required for manufacture of one article of the finished product and the rate per hour are as under:<br><br>The Actual production in a month was 100 articles.<br>The details of actual hours worked by the different labour categories and the rate per labour hour are as under:<br><br>Calculate the following labour variances from the following data:<br>Labour rate variance<br>Labour efficiency variance<br>Labour cost variance | [12] |

| Category of labour | Labour hours per unit | Rate per labour hour |
|---|---|---|
| Skilled | 10 | 100 |
| Semi Skilled | 7 | 50 |
| Unskilled | 12 | 20 |

| Category of labour | Labour hours for 100 units | Rate per labour hour |
|---|---|---|
| Skilled | 1020 | 110 |
| Semi Skilled | 690 | 55 |
| Unskilled | 1220 | 19 |